Lecture 5 · Production I: PMP and Aggregation

MWG Ch. 5.A–5.C · the consumer's theory, minus the budget constraint

Advanced Microeconomics 1

Monday 12 October

The two goals, firm edition

Same frame, new agent

(a) What do firms actually do — is the maximization hypothesis true of them? **(b)** Is production efficient — and what would we gain by fixing it?

- For firms, (a) splits into two separable claims — they **optimize**, and they **take prices** — and data can accept the first while rejecting the second. Hold that thought; it organizes both lectures.
- **The ladder, descending:** price-taking + profit max buys everything (today); drop price-taking and *cost minimization* survives (Thursday); drop optimization and only the technology remains — and Thursday's evidence will ask whether even that rung holds.
- The puzzle the block answers, stated now: within narrow US industries the 90th-percentile plant gets twice the output of the 10th from the same measured inputs; in India and China, $\sim 5\times$ (Syverson, JEL 2011 — **this block's (a) paper**). Under the full hypothesis that dispersion shouldn't survive competition. Where does it come from?

Why production theory is (mostly) easier

- The objective is **observable and cardinal**: profit, in currency. No ordinal hedging, no monotone-transformation caveats.
- There is **no budget constraint** $\Rightarrow$ no wealth effects $\Rightarrow$ no Giffen goods, no compensation subtleties, and — the big one — aggregation works (today's finale). Goal (a) is *easier* to test on firms than on consumers.
- The whole duality toolkit from Ch. 3 transfers wholesale; you already know today's proofs, just with new labels.
- Where it gets subtle again: *who* is the firm and *why* profit? (5.G, Thursday) — which is exactly where Thursday's evidence lands the hardest blow.

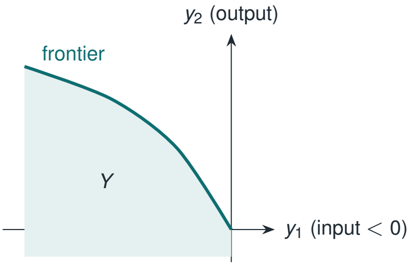
Production sets

Netputs

Definition · production vector and production set

A **production vector (netput)** is $y \in \mathbb{R}^L$: $y_\ell > 0$ means output of good ℓ, $y_\ell < 0$ means input. The **production set** $Y \subseteq \mathbb{R}^L$ collects all technologically feasible y.

- One convention, big dividends: profit is simply $p \cdot y$ — revenues and costs in one inner product; input demand and output supply become one object.
- Sign errors here are the single most common exam disaster in Ch. 5. Drill it now.



A menu of properties (MWG 5.B)

Property	Content
Y nonempty, closed	technical housekeeping (limits of feasible plans are feasible)
No free lunch	$Y \cap \mathbb{R}^L_+ \subseteq \{0\}$: no outputs without inputs
Possibility of inaction	$0 \in Y$: can shut down (sunk costs violate this — timing matters)
Free disposal	$y \in Y,\ y' \leq y \Rightarrow y' \in Y$: can absorb extra inputs harmlessly
Irreversibility	$y \in Y,\ y \neq 0 \Rightarrow -y \notin Y$: can't run production backwards
Nonincreasing RTS	$y \in Y \Rightarrow \alpha y \in Y$ for all $\alpha \in [0,1]$ (scale *down*)
Nondecreasing RTS	$y \in Y \Rightarrow \alpha y \in Y$ for all $\alpha \geq 1$ (scale *up*)
Constant RTS	both: Y is a cone
Additivity	$y, y' \in Y \Rightarrow y + y' \in Y$ (replication; entry)
Convexity	$y, y' \in Y \Rightarrow \alpha y + (1-\alpha) y' \in Y$

- These are à la carte, not a package. Each theorem today states which it needs — keep the ledger.

How the properties interlock

Proposition · relations worth proving once

1. Convexity + possibility of inaction $\Rightarrow$ nonincreasing returns to scale.
   (Scale down = mix y with 0.)
2. Y is a **convex cone** $\iff$ constant returns (and then additivity holds too).
3. Additivity + nonincreasing RTS $\Rightarrow$ Y is a convex cone. (EC2, Problem 1)

- Reading of (3): if technologies can be *replicated* (additivity across plants/firms), then at the industry level, decreasing returns cannot survive — the aggregate looks CRS. A first hint that "the industry" is better behaved than "the firm."
- Nonconvexities (setup costs, indivisibilities) are realistic and important — they break today's supporting-price logic; we meet the wreckage in Thursday's efficiency discussion.

Describing the frontier

- **Transformation function**: $Y = \{y : F(y) \leq 0\}$, frontier $F(y) = 0$. Marginal rate of transformation:

$$\mathrm{MRT}_{\ell k}(y) = \frac{\partial F/\partial y_\ell}{\partial F/\partial y_k} \quad = \text{slope of the frontier.}$$

- **Single output**: $q = f(z)$, inputs $z \in \mathbb{R}_+^{L-1}$; $Y = \{(-z, q) : q \leq f(z)\}$. Isoquants; marginal rate of technical substitution $\mathrm{MRTS}_{\ell k} = f_\ell / f_k$.
- Returns to scale for f: $f(\alpha z) \gtreqless \alpha f(z)$. Cobb–Douglas $f(z) = z_1^\alpha z_2^\beta$: RTS governed by $\alpha + \beta$ vs. 1 — the running example all week.

The profit maximization problem

PMP

PMP

Given $p \gg 0$ and Y (nonempty, closed, free disposal):

$$\pi(p) = \max_{y \in Y} \; p \cdot y, \qquad y(p) = \{y \in Y : p \cdot y = \pi(p)\}.$$

- π: **profit function**; $y(p)$: **supply correspondence** — outputs supplied and inputs demanded, one object.
- Single-output version: $\max_z \; pf(z) - w \cdot z$; FOC $p\, f_\ell(z^*) \le w_\ell$ (equality if $z^*_\ell > 0$): *marginal revenue product* = *input price*.

Existence pathology

With nondecreasing returns to scale, either $\pi(p) = 0$ or $\pi(p) = +\infty$: any profitable plan can be scaled up forever. Under CRS, active production requires exactly zero profit — the knife-edge that GE will exploit, and the reason Thursday rebuilds the theory on *cost* instead.

-

Properties of π and y

Proposition · MWG 5.C.1

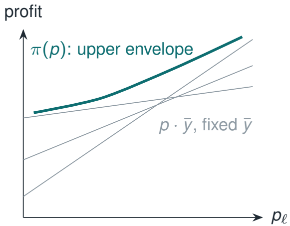
For Y closed with free disposal:

1. π is **homogeneous of degree 1**; y is **h.d.0**.
2. π is **convex** in p.
3. If Y convex: $y(p)$ convex-valued (and $Y = \{y : p \cdot y \leq \pi(p)\ \forall p \gg 0\}$ — duality: π encodes all of Y).
4. **Hotelling's lemma**: if $y(\bar{p})$ is a singleton, π is differentiable at $\bar{p}$ and $\nabla\pi(\bar{p}) = y(\bar{p})$.
5. If y is differentiable: $Dy(\bar{p}) = D^2\pi(\bar{p})$ is **symmetric and positive semidefinite**, with $Dy(\bar{p})\,\bar{p} = 0$.

- Compare line by line with $e(p,u)$ and $h(p,u)$: same theorem, mirror signs. Min problem $\to$ concave value, NSD derivative matrix; max problem $\to$ convex value, PSD derivative matrix.

Hotelling's lemma: trick #2 of 3

- *Envelope proof*: p_ℓ enters the objective $p \cdot y$ linearly with coefficient y_ℓ; at the optimum, $\partial\pi/\partial p_\ell = y_\ell(p)$. $\square$
- *Convexity proof*: for each fixed plan $\bar{y}$, $p \mapsto p \cdot \bar{y}$ is linear; π is the pointwise **max** — an *upper* envelope of linear functions, hence convex.
- Economics of convexity: a passive firm has linear profits in prices; *re-optimization* tilts the true π above every tangent. Firms like price variability (Oi 1961).



The trilogy so far

$e(p,u)$: min $\Rightarrow$ concave, $\nabla_p e = h$. $\pi(p)$: max $\Rightarrow$ convex, $\nabla\pi = y$. One more member joins in L7.

The law of supply

Proposition · law of supply

For any p, p' and $y \in y(p)$, $y' \in y(p')$:

$$(p - p') \cdot (y - y') \;\geq\; 0.$$

Proof. Optimality at p: $p \cdot y \geq p \cdot y'$. Optimality at p': $p' \cdot y' \geq p' \cdot y$. Add and rearrange. $\square$

- Own-price version: output up when its price rises; input demand down when its wage rises. No compensation required — contrast Ch. 2/3, where the law of demand held only for *compensated* changes.
- Why the asymmetry: no budget constraint $\Rightarrow$ no income effects to fight the substitution effect. Quantities respond to prices the "intuitive" way, unconditionally. This two-line proof + one-line explanation is a prelim classic.
- **Goal (a), firm edition, both regimes:** a supply *function* is rational iff $Dy = D^2\pi$ is symmetric PSD — no compensation needed, because there is nothing to compensate. Finite data: the firm's GARP is **WAPM** (Varian 1984, in passing): $p^t y^t \geq p^t y^s$ for all s, t. Easier than the consumer's test in every way.

Worked example: one input, $f(z) = z^\alpha$, $\alpha \in (0,1)$

Solve $\max_{z \geq 0} \; pz^\alpha - wz$:

$$z(p,w) = \left(\frac{\alpha p}{w}\right)^{\frac{1}{1-\alpha}}, \qquad q(p,w) = \left(\frac{\alpha p}{w}\right)^{\frac{\alpha}{1-\alpha}}, \qquad \pi(p,w) = (1-\alpha)\, p^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{w}\right)^{\frac{\alpha}{1-\alpha}}.$$

- Verify on the spot: π h.d.1 in (p, w); $\partial\pi/\partial p = q$ and $\partial\pi/\partial w = -z$ (Hotelling, both signs); π convex in p.
- Watch $\alpha \to 1$ (CRS): $z, q, \pi \to$ the $0/\infty$ pathology, exactly as advertised.
- PS2 does the two-input version with $\alpha + \beta < 1$ and makes you check every property in 5.C.1.

Aggregation (5.E) — and the Ch. 4 cameo

What we skipped: aggregate demand is lawless

- Ch. 4's question: does *aggregate* demand $x(p) = \sum_i x_i(p, w_i)$ inherit the structure of individual demand?
- **Sonnenschein–Mantel–Debreu** (1972–74), informally: on any compact set of prices with $p \gg 0$, *any* continuous function satisfying homogeneity and Walras' law is the aggregate excess demand of *some* economy of well-behaved rational consumers.
- Translation: individual rationality imposes essentially no restrictions on aggregate demand — WARP can fail, Slutsky structure evaporates, multiple equilibria roam free. Wealth effects, distributed across heterogeneous agents, wash the structure out.
- Escape hatch: **Gorman form** — $v_i(p, w_i) = a_i(p) + b(p)w_i$, identical $b(\cdot)$ — makes aggregate demand independent of the wealth distribution, resurrecting a representative consumer. You prove it in EC2; modern heterogeneous-agent macro (HANK) exists precisely because Gorman is a heroic assumption.

...but aggregate supply inherits everything

Proposition · MWG 5.E.1

For firms $Y_1, \dots, Y_J$ with aggregate set $Y = \sum_j Y_j$ and $p \gg 0$:

$$\pi(p) = \sum_j \pi_j(p) \qquad \text{and} \qquad y(p) = \sum_j y_j(p),$$

where π, y belong to the *aggregate* set Y.

Proof idea. Sums of individually optimal plans are feasible for Y and earn $\sum_j \pi_j(p)$; no aggregate plan can beat that, since any $y = \sum_j y_j$ is weakly dominated by letting each firm optimize its own component. Conversely, decompose any aggregate optimum: each component must be individually optimal, or a within-firm swap raises total profit. $\square$

- Corollary (the **representative producer**): the industry behaves *exactly* as one price-taking firm running the pooled technology — decentralized profit maximization $=$ centralized. No assumption on the distribution of anything. The law of supply aggregates too: sums of PSD matrices are PSD.

The scoreboard

	Demand	Supply
adds up?	yes	yes
structure survives?	**no** (SMD)	**yes** (5.E.1)
representative agent	needs Gorman	free

- One cause: no wealth effects on the production side. Firms have no budget constraints, so reallocating "wealth" across firms has nothing to undo.
- "The representative firm" is a theorem; "the representative consumer" is a hope.

Scorecard, takeaways, readings

Takeaways

- **Goal (a), best case:** with price-taking + profit max, the test is unconditional — π convex, $\nabla\pi = y$ (Hotelling), *Dy* symmetric PSD, the law of supply with *no compensation*. Consumer duality with the signs flipped and the income effects deleted.
- **Goal (a) scales:** supply aggregates perfectly; demand (SMD) does not. No wealth effects is the whole story — "the representative firm" is a theorem, "the representative consumer" a hope.
- **The puzzle stands:** Syverson's dispersion is untouched by today. Thursday walks down the ladder to find it — and prices it.

Required: MWG 5.A–5.C (through profit maximization) and 5.E.
The block's papers: *(a)* Syverson (JEL 2011), Secs. 1–2 now. *(b)* Hsieh–Klenow (QJE 2009) — skim the intro before Thursday. *The third is revealed Thursday.*
In passing: Kirman (1992); Varian (1984); MWG 4.C for the honest SMD statement.
Thursday: cost minimization — the rung that survives — then the dispersion puzzle, decomposed and priced.